

4.10.24

EE25BTECH11022 - sankeerthan

Question: Find the equations of the lines through the point of intersection of the line $x - y + 1 = 0$ and $2x - 3y + 5 = 0$ and whose distance from the point $(3, 2)$ is $\frac{7}{5}$.

Solution: Given lines are

$$(1 \quad -1) \begin{pmatrix} x \\ y \end{pmatrix} + 1 = 0, (2 \quad -3) \begin{pmatrix} x \\ y \end{pmatrix} + 5 = 0 \quad (1)$$

$$\begin{pmatrix} 1 & -1 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix} \quad (2)$$

Inverse of coefficient matrix

$$\det = 1 \times (-3) - (-1) \times 2 = -3 + 2 = -1 \quad (3)$$

$$A^{-1} = \frac{1}{-1} \begin{pmatrix} -3 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 2 & -1 \end{pmatrix} \quad (4)$$

Intersection point of given two lines is

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = A^{-1} \begin{pmatrix} -1 \\ -5 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad (5)$$

General line through (x_0, y_0) with normal vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is

$$(a \quad b) \left(\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \quad (6)$$

Distance from point $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ to line is

$$Distance = \frac{|(a \quad b) \left(\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right)|}{\sqrt{(a \quad b) \begin{pmatrix} a \\ b \end{pmatrix}}} = \frac{7}{5} \quad (7)$$

$$\frac{|(a \quad b) \left(\begin{pmatrix} 3-2 \\ 2-3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right)|}{\sqrt{(a \quad b) \begin{pmatrix} a \\ b \end{pmatrix}}} = \frac{7}{5} \quad (8)$$

$$\frac{|a - b|}{\sqrt{a^2 + b^2}} = \frac{7}{5} \quad (9)$$

Let $t = \frac{a}{b}$ ($b \neq 0$)

$$\frac{|t-1|}{\sqrt{t^2+1}} = \frac{7}{5} \quad (10)$$

Squaring on both sides

$$25(t-1)^2 = 49(t^2+1) \quad (11)$$

$$25(t^2-2t+1) = 49t^2+49 \quad (12)$$

$$25t^2-50t+25 = 49t^2+49 \quad (13)$$

$$24t^2+50t+24 = 0 \quad (14)$$

$$12t^2+25t+12 = 0 \quad (15)$$

$$t = \frac{-25 \pm \sqrt{25^2 - 4 \times 12 \times 12}}{2 \times 12} = \frac{-25 \pm 7}{24} \quad (16)$$

Solutions $t_1 = -\frac{3}{4}, t_2 = -1$

Normal vectors ($let b = 1$) :

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (17)$$

Equation constants $c = -a \times 2 - b \times 3$

$$c_1 = (-2) \times \left(-\frac{3}{4}\right) - 3 \times 1 = \frac{3}{2} - 3 = -\frac{3}{2} \quad (18)$$

$$c_2 = (-2) \times (-1) - 3 \times 1 = 2 - 3 = -1 \quad (19)$$

required equations of line are

$$\begin{pmatrix} -\frac{3}{4} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - \frac{3}{2} = 0 \quad (20)$$

$$\begin{pmatrix} -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - 1 = 0 \quad (21)$$

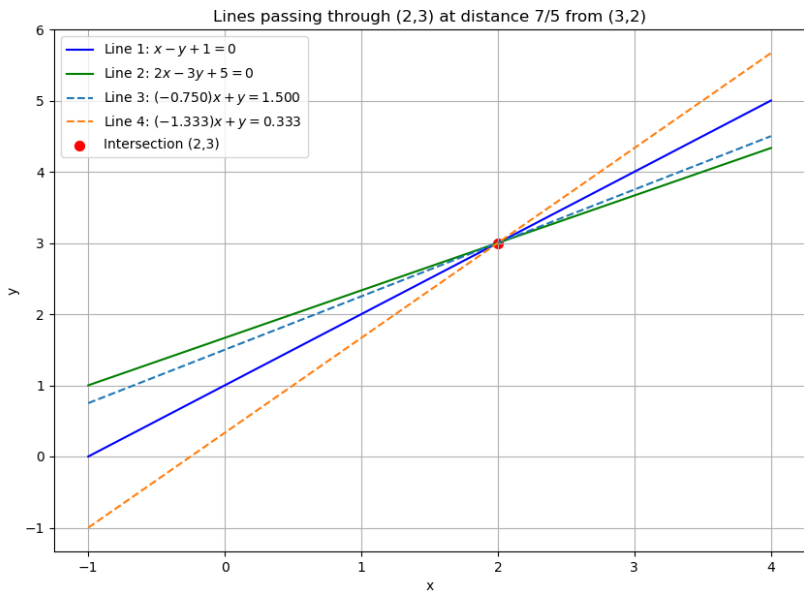


Fig. 0.1